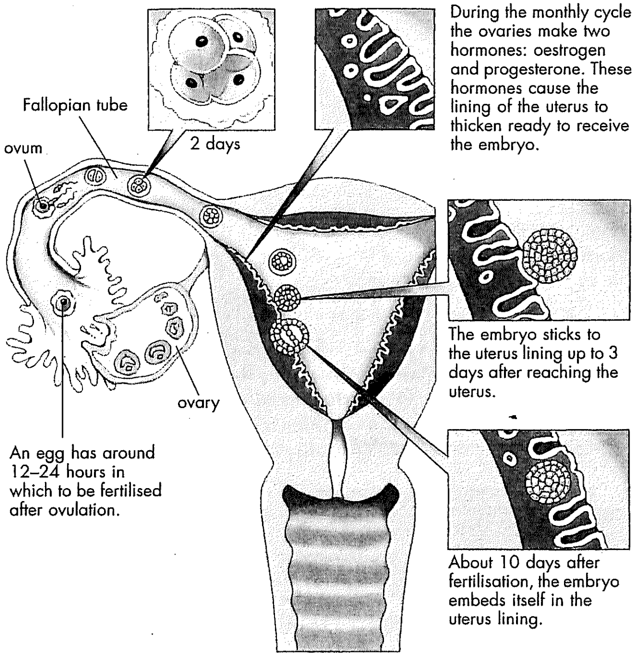
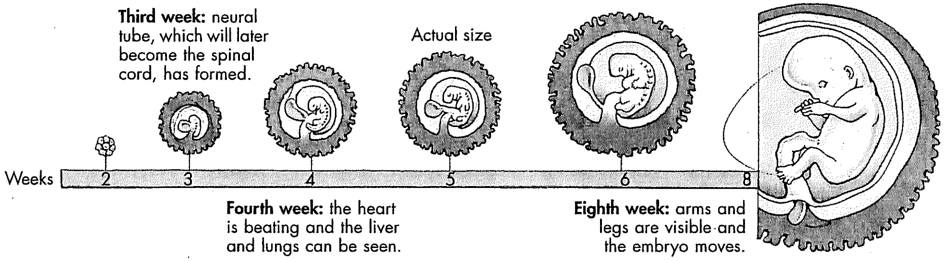
10.6 Making babies

Multiplication is the name of the reproductive game. Sometimes one and one can make three!



The *first eight* weeks

Conception occurs when the egg cell and sperm unite to form a zygote. When the zygote has divided into many more cells, it is known as an **embryo**. About ten days after fertilisation, the embryo completely embeds itself in the uterus lining (endometrium). This process is called **implantation**.

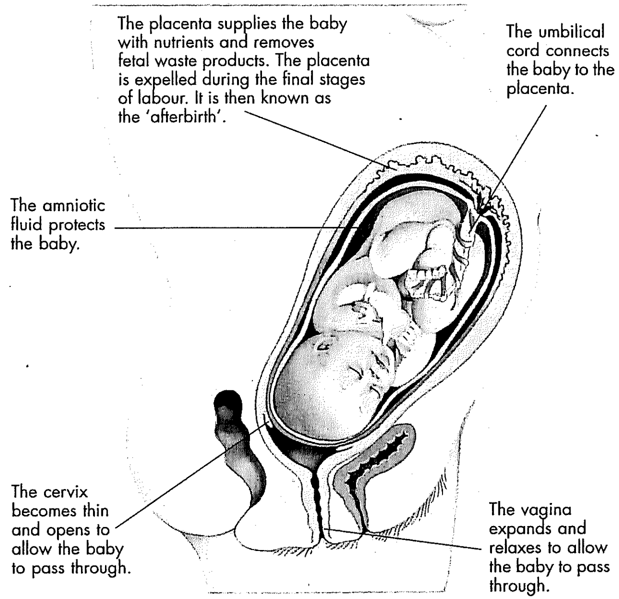


'After *eight*'

In humans, at about eight weeks, when the embryo has developed a distinct head, arms and legs, it is called a **fetus**. The fetus obtains its nutrients and oxygen through a special organ called the **placenta**. This organ is connected to the mother's blood vessels through the uterus. The placenta also absorbs fetal waste products and acts as a barrier against harmful substances. The unborn child continues to develop inside a sac which is filled with fluid (called amniotic fluid) for the rest of its time within the uterus. The total time spent in the uterus is often called the **gestation period**. In humans this is usually about 40 weeks. If a baby is born before 37 weeks it is called **premature** and usually requires extra care and assistance.

Approximate size of a fetus at different stages of development

Development (weeks)	Length (cm)	Mass (g)
8	3	2
12	7.5	18
16	16	140
40	51	3400



Ready for birth — the baby at 40 weeks gestation

Giving *birth*

There are three stages involved in giving birth to a baby. Quite often this is referred to as **labour** because it can be a lot of hard work for the mother. During the first stage the cervix gradually widens. In the second stage, the woman feels a strong urge to push with each contraction of the uterus. It is during this stage that the baby is born through the vagina, or birth canal. Usually the baby is born head first. Sometimes, however, the baby is born feet first; this is referred to as a **breech** birth and is often more difficult. The third stage lasts from the delivery of the baby until the placenta is delivered.

In some cases the baby or the mother need extra assistance. A **caesarean** operation may be performed, in which doctors surgically remove the baby by cutting through the mother's abdomen to her uterus.

The *same* or *different*

When more than one fetus is present in the uterus, it is called a multiple pregnancy. **Identical twins** are produced after a sperm and an egg have fused together, and the embryo has separated at an early stage of development so that the two offspring are identical. Because the offspring contain the same genetic mix, they are always of the same gender (i.e. both girls or both boys). **Fraternal twins**, however, are the result of two different pairs of gametes joining. Each fraternal twin has been formed from a different sperm and a different egg.

Activities

Remember

1. State the differences between a zygote, an embryo and a fetus.
2. List the following in the correct order:
   birth, fertilisation, ovulation, growth, implantation.
3. In which part of the female reproductive system does the fetus develop?
4. How is a breech birth different from a caesarean?
5. What is the difference between:
   (a) fertilisation and implantation
   (b) fraternal and identical twins?

Using data

Construct a graph, using information in the table and the illustrations, to show the changes in length of the embryo from 2 to 8 weeks and the length and weight of the fetus from 8 to 40 weeks.

Create

Make scale models of the fetus at each age shown in the table.

Think

1. Correctly match the key events:

Day	Key event
14	implantation
15	first cell division
16	fertilisation
23	ovulation

2. Suggest why menstruation must stop during pregnancy. What would happen if this was not the case?

Research

1. Research and report on one of the following:
   endometriosis, prolapse of the uterus, cervical cancer, hysterectomy, ectopic pregnancy.
2. Research and report on one of the following antenatal tests:
   ultrasound scanning, amniocentesis, chorionic villus sampling.

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Pregnancy and the placenta

The period of development between fertilization and birth is called **pregnancy**. Human pregnancy lasts about nine months. During this time hormones in the mother's blood prevent the monthly occurrence of menstruation and ovulation. Implantation of a developing embryo is the signal which ensures that these hormones flow.

If implantation takes place, cells surrounding the embryo produce hormones which prevent breakdown of the corpus luteum so that it continues producing progesterone. This hormone stimulates further growth of the uterus and milk-producing tissues in the breasts.

A developing embryo must have oxygen and food. At first it obtains these directly from its mother's blood flowing through vessels in the lining of the uterus. Within three weeks of implantation, however, food and oxygen are absorbed within an organ called a **placenta**.

A placenta is a disc-shaped organ with millions of tiny root-like outgrowths called **villi**. These villi are embedded in the lining of the uterus (Figs. 1 and 2). A placenta is connected with the embryo by a tube called the **umbilical cord**. The embryo's heart pumps blood through blood vessels in the umbilical cord into the placenta where it flows through capillaries in the placental villi (Fig. 2). The villi hang inside spaces in the uterus wall filled with the mother's blood. In these spaces food and oxygen pass from the mother's blood into the embryo's blood, while at the same time carbon dioxide and other wastes pass from the embryo's blood into the mother's blood.

The placenta is a temporary endocrine organ. It produces large amounts of oestrogen and progesterone. The oestrogen prevents the pituitary gland from making follicle-stimulating hormone so that follicles do not develop during pregnancy. The progesterone ensures that the uterus grows at the same rate as the baby and also that the breasts are ready to produce milk soon after the baby is born.

The baby develops inside a bag of liquid called the **amnion** (Figs. 1 and 2). The amnion acts as a shock-absorber—the liquid in it cushions the baby against jolts and knocks as the mother moves about. It also helps prevent the baby being injured if anything hits or presses against the mother's body.

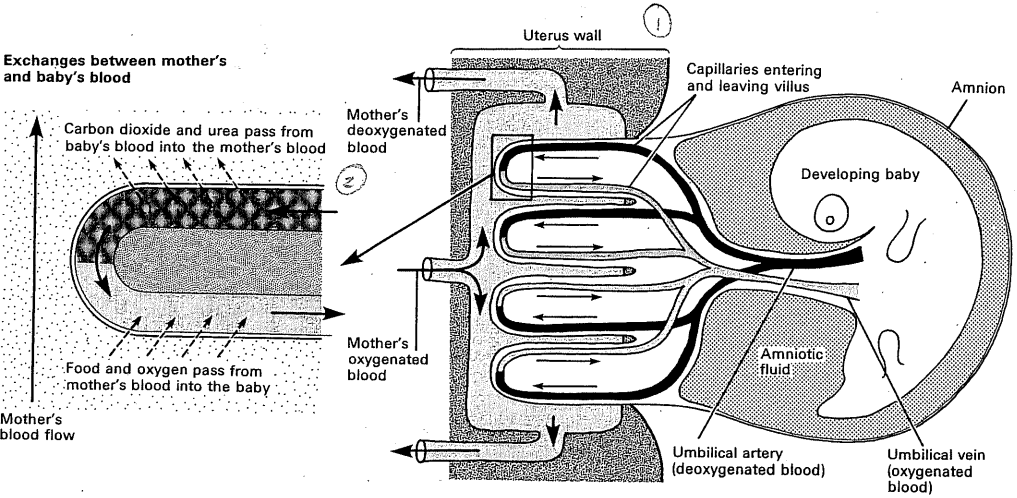

Fig. 1. Diagram of an embryo and its placenta. The embryo's heart pumps blood through the placenta where it passes through capillaries in placental villi. The villi are in blood-filled spaces in the uterus wall. Food and oxygen pass from the mother's to the embryo's blood while carbon dioxide and urea pass from the embryo's to the mother's blood.

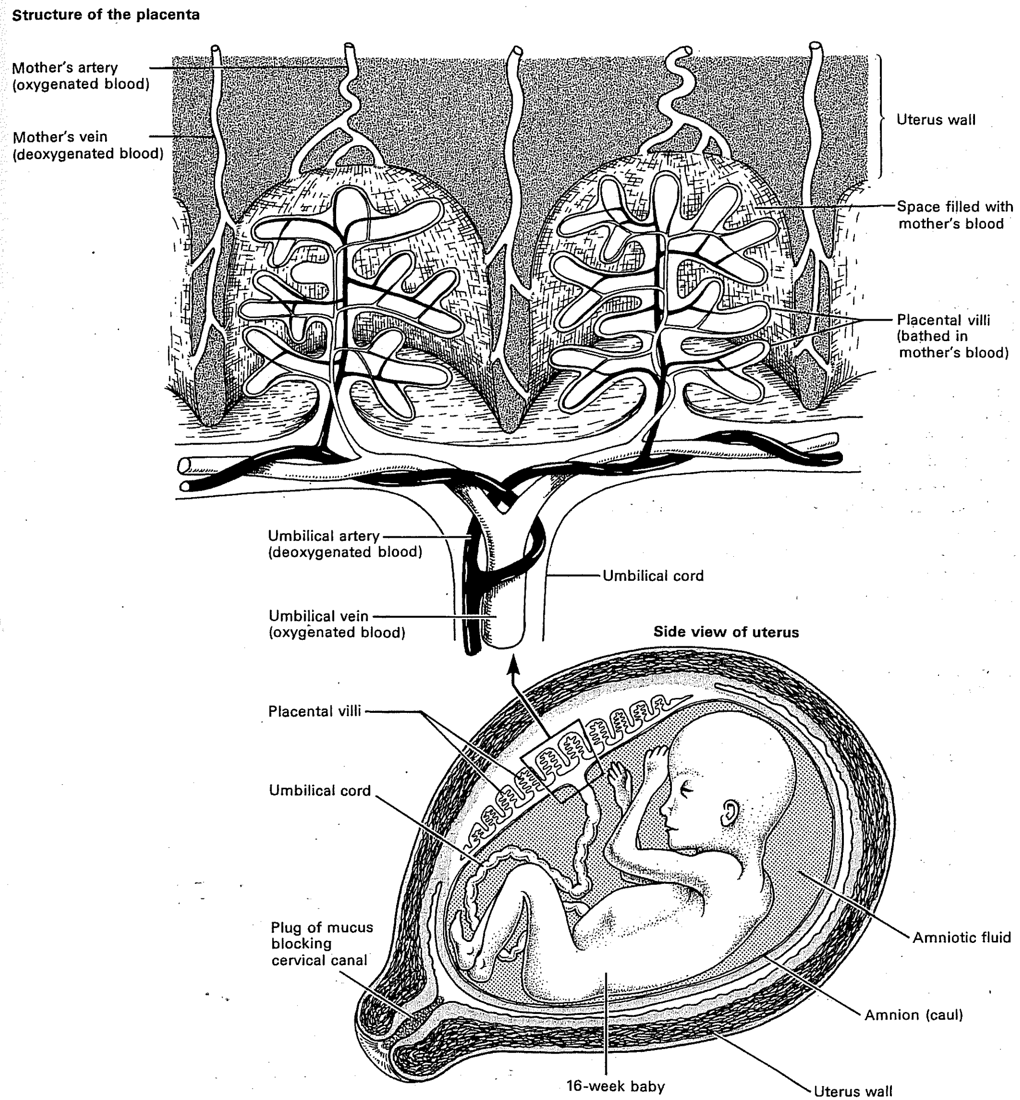

Fig. 2. Structure of the placenta.

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Development before birth

Ideally the age of an unborn baby should be calculated from the moment of fertilization. But this is difficult because it is not always possible to know exactly when fertilization occurred. To overcome this difficulty it is usual to calculate a developing baby's age from the first day of the mother's last period, since this date can be calculated with some certainty. This system gives the **menstrual age** of the baby, and is used in the following descriptions. It must be remembered, however, that the menstrual age is always about two weeks too much.

Week 5

A fertilized ovum is smaller than a printed full stop, but by week 5 (about three weeks after fertilization) it is about 5 mm long. It has a bulge which will soon be a head; the brain, eyes, and ears are beginning to develop; and a simple heart pumps blood through a small placenta. The embryo has a long tail and small lumps on its sides are forming into arms and legs.

Weeks 8 to 10

During this period the embryo becomes clearly human in appearance and from now until birth is called a **foetus**. It has a face, and limbs with fingers and toes. Most of its internal organs are fully formed and will continue to grow until birth.

Week 20

The foetus is now about 25 cm long. The mother can feel it moving occasionally in her womb and its heart-beats can be heard with a stethoscope. Sex organs are developed to a stage at which differences between a boy and girl are obvious.

Week 28

By this time the foetus is 35 to 38 cm long. It has soft downy hair and eye-lashes, and can open its eyes. Finger and toe nails have hardened and milk teeth are developing in its jaws. Its body is covered with white grease called **vernix** which stops the skin becoming waterlogged with amniotic fluid. Its organs have now grown to a stage at which the baby is just able to survive with expert care if removed from its mother. In technical terms it is **legally viable**.

Week 40

The pregnancy is now **at term**—the baby is ready to be born. It weighs between 3 and 3.5 kg and is about 50 cm long.

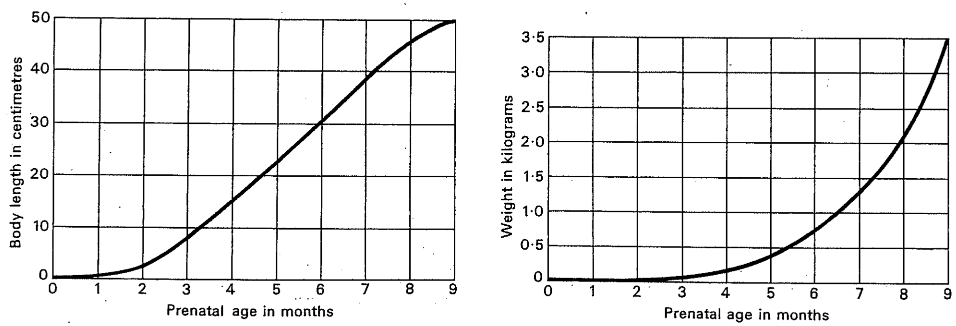
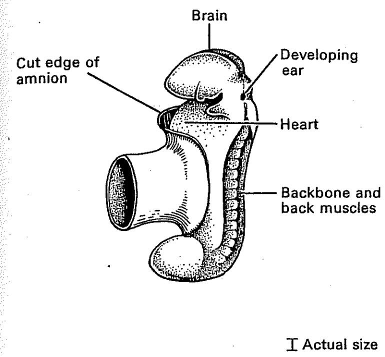
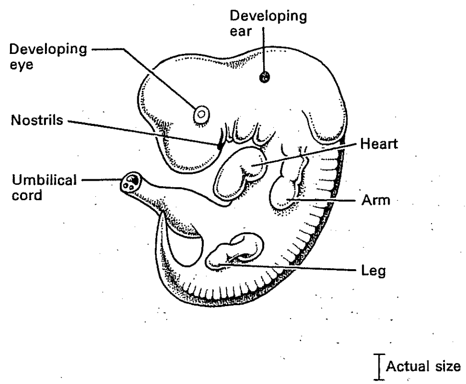


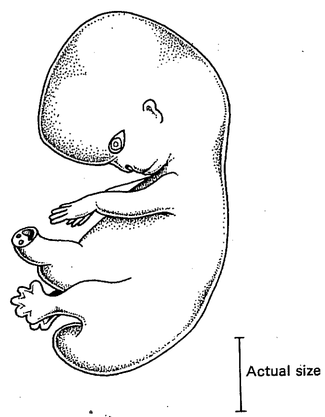
Fig. 1. Changes in the length and weight of a baby up to birth (prenatal period).



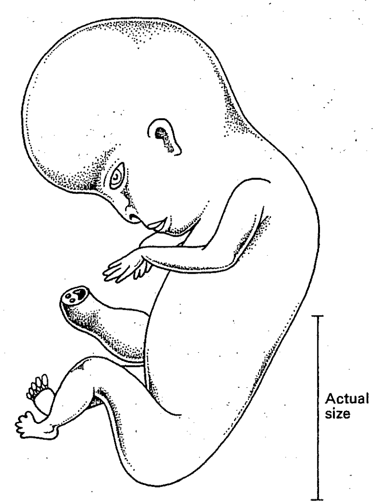
Week 3 The embryo is about 2·0 mm long. Already the developing brain, ears, heart, and backbone can be seen



Weeks 4 to 5 The embryo is about 5 mm long. Its brain, eyes, and ears are developing and a heart pumps blood to a placenta. Limbs are represented by lumps on each side of the body



Weeks 6 to 7 The embryo is about 14 mm long. Its limbs well-formed. Nose and mouth are developing and lungs, liver, and kidneys are taking shape



Weeks 8 to 9 The body is about 35 mm long. It is now called a foetus and is human in appearance. The body is almost completely formed and will continue growth until birth

Fig. 2. Development of a baby inside the uterus.

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Birth

During the months before birth the uterus walls develop a thick layer of muscle. Birth begins when hormones in the mother's blood cause this muscle to start contracting rhythmically. At first the contractions are weak and far apart but gradually become stronger and closer together and cause **labour pains**. Eventually the contractions push the baby out of the mother's body.

A few weeks before birth most babies turn within the uterus until their head points downwards towards the cervix (Fig. 2A). Birth occurs in three stages. First the cervix expands allowing the baby's head to pass into the vagina (Fig. 2B). Second, contractions push the baby through the vagina and out of the mother's body. This is called **delivery** (Fig. 2C). Third, further contractions push the placenta, or **after-birth**, out of the mother's body (Fig. 2D).

After leaving the mother a baby takes its first breath and usually begins crying. Crying is a reflex action which establishes regular breathing. These events bring about changes in the baby's circulatory system.

Circulatory changes at birth

A foetus obtains oxygen from its placenta. Its lungs do not function and receive very little blood. But at birth the placenta stops functioning and its blood supply is cut off. There is then a rapid alteration in circulation so that blood is diverted to the lungs as they inflate with air for the first time.

Figure 1 shows how this happens. In a foetus blood entering the right atrium is not pumped to the lungs; it by-passes them in two ways. Some blood passes from the right atrium to the left atrium through a hole called the **foramen ovale**, and blood which enters the pulmonary artery is transferred to the aorta by a vessel called the **ductus arteriosus**.

At birth blood supply to the placenta is cut off and some blood begins flowing to the lungs. When this blood returns to the heart it raises blood pressure in the left atrium which presses a flap of tissue over the foramen ovale sealing it up so that blood no longer crosses from the right to the left atrium (Fig. 1). A few hours later the ductus arteriosus closes making all the blood pumped from the right ventricle flow to the lungs.

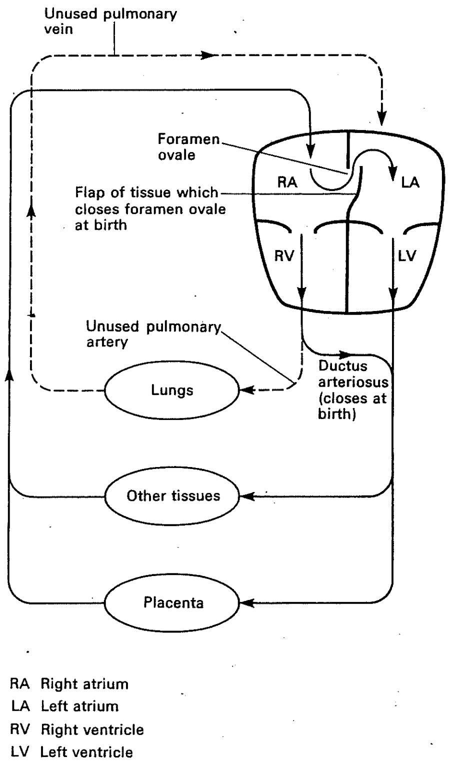
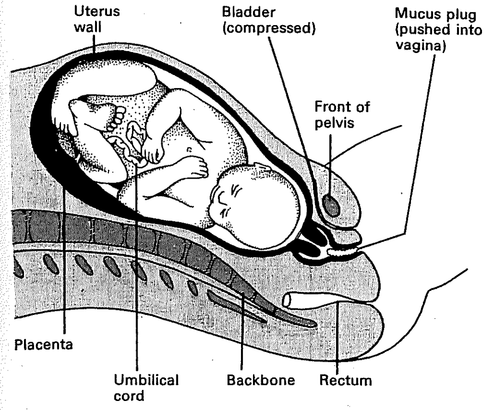
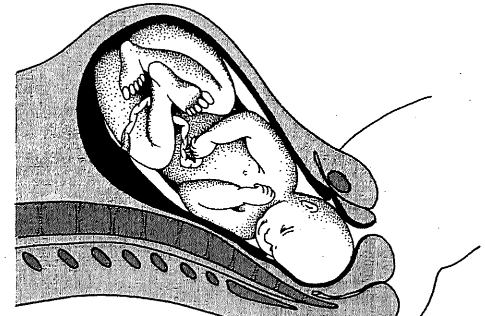


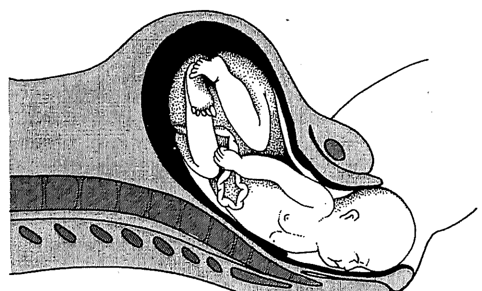
Fig. 1. Changes in circulation at birth. In a foetus blood entering the right atrium is diverted from the lungs in two ways. Some blood passes through the foramen ovale into the left atrium, and the remainder is diverted from the pulmonary artery through the ductus arteriosus into the aorta. At birth the foramen ovale and ductus arteriosus close up so that blood is pumped from the right ventricle to the lungs.



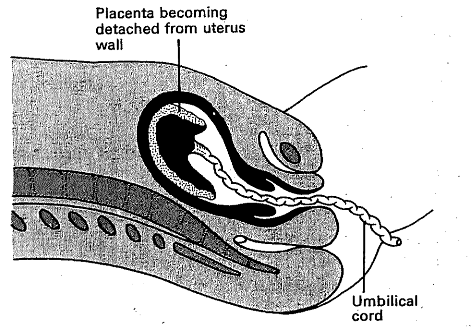
A A few days before birth the baby turns in the uterus until its head points towards the cervix. Birth begins when the uterus starts to contract rhythmically



B During the first stage of birth contractions force the head onto the cervix stretching it until the head passes through to the vagina. The mucus plug is displaced and the amnion usually bursts



C During the second stage of birth contractions are strong and close together. The baby moves to a face-down position and is pushed out of the mother's body



D During the third stage of birth further contractions of the uterus push the placenta and umbilical cord (after-birth) out of the mother's body

Fig. 2. The birth of a baby.

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The newborn baby

A newborn baby displays several reflex responses to changes in its surroundings (Fig. 1). Many of these reflexes are used to check the baby's health and the condition of its nervous system.

Feeding

Both breast feeding and bottle feeding produce normal healthy babies.

Breast feeding Most mothers are able to breast feed their babies, and if they can this method has a number of advantages over bottle feeding. Breast milk costs nothing and does not have to be prepared. It is available immediately, which is convenient, especially for the 2 a.m. feeds. Breast milk is always fresh and, provided the nipples are clean, free from germs. It contains ideal proportions of all the foods a baby requires and is digested rapidly and easily. There is evidence that, compared with bottle-fed babies, those which are breast-fed are less likely to have digestive upsets, skin disorders (including nappy rash), and respiratory infections.

Babies are usually put to the breast within eighteen hours of birth. Some hospitals, however, believe that this should be done as soon as possible after delivery because the sucking of a baby at a breast produces a reflex response in the mother which causes her pituitary gland to release the hormones **prolactin** and **oxytocin**. Prolactin stimulates milk production in the breasts and oxytocin causes the breasts to eject milk. Oxytocin also causes the uterus to contract to its former size.

When a baby is first put to the breast it receives a thick liquid called **colostrum**, which is particularly rich in proteins. True milk does not appear until two or three days after birth but during this time a baby has little appetite, sleeps most of the time, and usually loses some weight. By the time true milk appears most babies have become more wakeful and hungry. The baby sucking at the breast stimulates production of milk so that the breasts adjust to the baby's demands, gradually increasing milk production as the baby grows and wants more.

Bottle feeding All artificial or formula milks for use in bottle feeding are based on cows' milk with various sugars and other substances added to make them more like human milk. Table 4 gives the differences between cow and human milk. The main advantages of bottle feeding are that the baby's food intake can be measured and that people other than the mother can help with feeding. The main disadvantages of bottle feeding are that formula milk is expensive and, although it is a good imitation of breast milk, it is not so easily digested. Furthermore, unless carefully cleaned and sterilized a bottle can transmit germs to a baby.

Care of mother and baby

Mother Soon after delivery a mother should carry out regular exercises to strengthen the muscles stretched out of shape during birth.

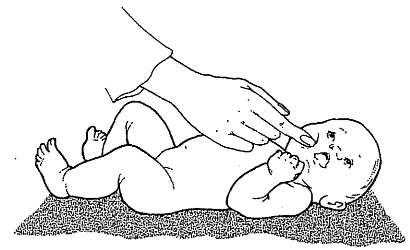
The large wound in the uterus wall where the placenta was attached takes four to six weeks to heal and during this time a substance called **lochia** is discharged from the wound and passes out of the vulva. The vulva must be kept extremely clean because, until the uterus heals, the reproductive and urinary passages can easily become infected with germs. Regular bathing is essential, preferably in water to which a little salt has been added.

Baby It takes a few weeks before a baby's temperature-regulating mechanism begins to function efficiently and before it builds up a layer of fat beneath its skin. At first, therefore, a baby's room should be heated to about 20°C. The air in a heated house tends to be very dry and this can damage the delicate membranes lining a baby's respiratory passages. So it is beneficial to take a baby out of doors in warm clothing for an hour or two a day.

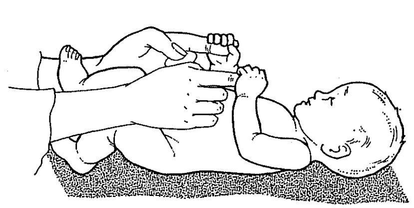
A baby should be bathed once a day, preferably before a meal. The mother should wash her hands before and after feeding her baby. If bottles are used they should be cleaned in water and detergent with a bottle brush and placed in sterilizing liquid. Teats should be boiled after use.

Table 4 Human and cows' milk compared

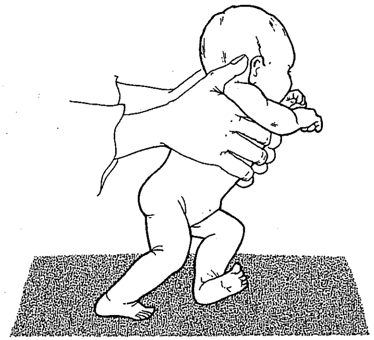
	Human *g per 100 g*	*Cow* *g per 100 g*
Water	88.0	88.0
Protein	1.2	3.3
Fat	3.5	3.6
Lactose	6.5	4.7
Vitamin C	4 to 8 mg	0.7 to 3 mg



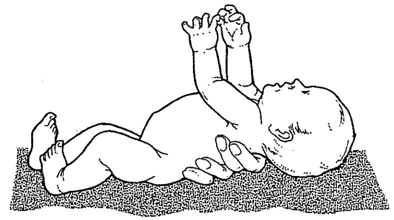
A Rooting reflex If either cheek is touched, the baby turns its head in the direction of the touch. This enables it to find the nipple of its mother's breast and, together with the sucking reflex, enables it to obtain food.



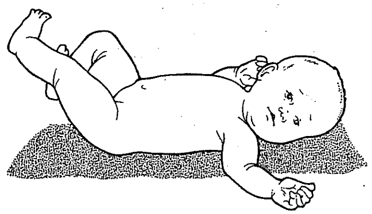
B Grasp reflex A baby will automatically grasp an object placed in its palm.



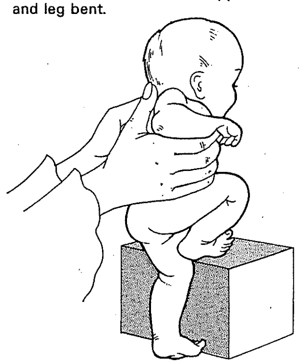
C Walking reflex If a baby is held with its feet touching the ground and moved forwards it will perform walking movements.



D Moro or startle reflex When startled a baby throws out its arms and legs, then pulls them back with the fingers curved, as if to catch hold of something.



E Resting position When resting on its back, a newborn baby lies with its head to one side, with the arm and leg on that side extended and the opposite arm and leg bent.



F Stepping reflex If a baby is held with the front of one leg in contact with an object, it will raise its other leg and step onto the object.

Fig. 1. Primitive reflexes in a newborn baby. These are used to check a baby's health and the condition of its nervous system. Within three months the primitive reflexes have been replaced by learned responses.

People involved in the birth process

Midwives	Obstetricians
Nurses	Doulas

6

Stages of birth

First stage
Second stage
Third stage

5. (i) List the changes which occur to the expectant mother's breasts early in her pregnancy.

____

____

(ii) Which hormones bring about these changes?

____

(iii) When is the hormone prolactin produced?

____

____

____

(iv) Where is the hormone prolactin released?

____

____

(v) What does prolactin do?

____

6. **Label** and **annotate** the diagram below.

i) ______________________

____

ii) ______________________

____

iii) ______________________

____

iv) ______________________

____

v) ______________________

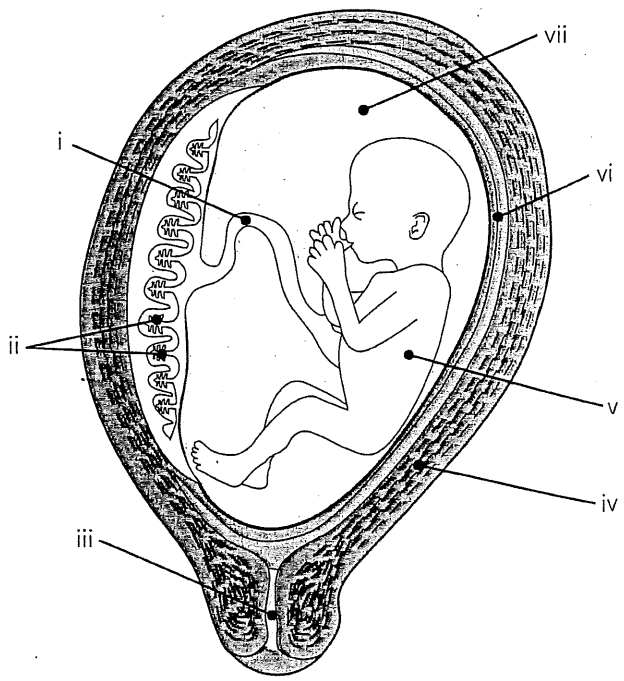
____

vi) ______________________

____

vii) ______________________

____



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the importance of each of the following:

(i) amniotic fluid

(ii) yolk sac

(iii) chorion

(i) Why is there a 40% increase in the volume of blood in a pregnant woman?

(ii) Why must she be particularly careful about what she eats and drinks?

(iii) Why is her intake most critical in the first 12 weeks of her pregnancy?

(i) What is a hormone?

(ii) Name and describe the type of gland that produces and releases a hormone.

(iii) How does a hormone reach its target?

Pregnancy Revision for test 2

Section II: Order of events in fetal Development: Place a number on the line in the order you think the events occur

11. ______ Cells begin to divide
12. ______ Fertilized egg implants into uterus wall
13. ______ Cervix dilates
14. _____ Pregnancy is full term baby turns head down
15. ______ Gender can be seen
16. ______ Fetus can hear
17. ______ Sperm cell enters egg
18. ______ All organs can function
19. ______ Heart begins to beat
20. ______ Hands and feet develop

Section III: Order of Development: Place these baby names in the order in which they occur during pregnancy by writing the number under each name. 1 being the first to occur and 5 being the last to occur.

A. Baby	B. Blastocyst	C. Zygote	D. Embryo	E. Fetus
22. ______	23. ______	24. ______	25. ______	26. ______

Section IV: Terms during pregnancy: Place the Terms with the correct definition for pregnancy.

Amniotic Sac	Umbilical Chord	Crowning
Placenta	Vernix	Lanugo

27. ________________ Supplies fetus with oxygen, supplies fetus with nutrients, and passes out wastes from the fetus.

28. ______________ Is a rope like structure that connects the embryo to the placenta. It has 1 large vein, and 2 arteries.

29. _______________ Acts as temperature control, protection from shock, barrier to infection.

30. _______________ Is fine downy hair that covers the fetus to help insulate, and regulate body temperature.

31. _______________ Is white, cheese-like substance that acts like a waterproof barrier against the amniotic fluid.

32. _______________ Is the appearance of the baby's head during delivery.

Section V: Order Signs of Labor and Stages of Birth: Place the stages of birth in the order in which they occur during pregnancy. 1 is the first to occur and 6 the last to occur.

Birth of Baby	Head appears	Birth of placenta
33. ______	34.______	35.______
Water Breaks	Contractions	Dilation of cervix
36. ______	37.______	38.______

Section VI: Match the correct type of pregnancy to the definition:

39. Natural Birth ______

40. Induced Labor ______

41. Breech Birth ______

42. Cesarean Section ______

43. Epidural______

44. Episiotomy______

45. Vacuum Extraction ______

46. Forceps______

A. Medication given or water sac broken by plastic hook (amnio-hook)

B. A shot administered in the woman's lower back to aid in pain relief.

C. A form of childbirth in which a surgical incision is made through a mother's abdomen and uterus to deliver one or more babies.

D. When baby delivered either foot first or buttocks first.

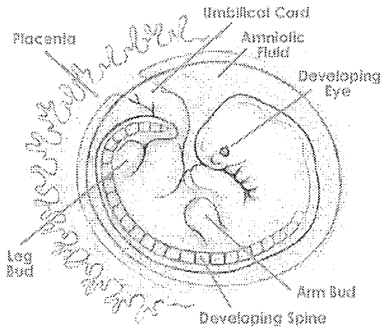
E. Delivery of a baby without using drugs or surgery during birth.

F. An incision between the vagina and anus (perineum) help with crowning of the baby, to prevent muscles from tearing

G. Assist mother if she becomes too tired, cup on baby's head with slight suction

H. Guide baby's head out of birth canal

Section VIII: Place the pictures in order from day one to delivery. Include trimester next to picture



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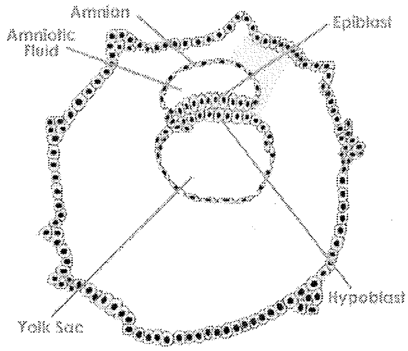

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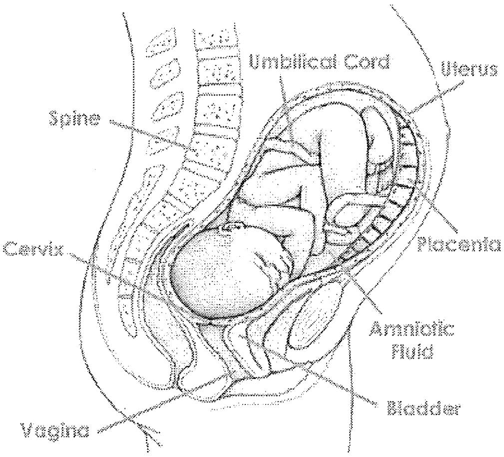

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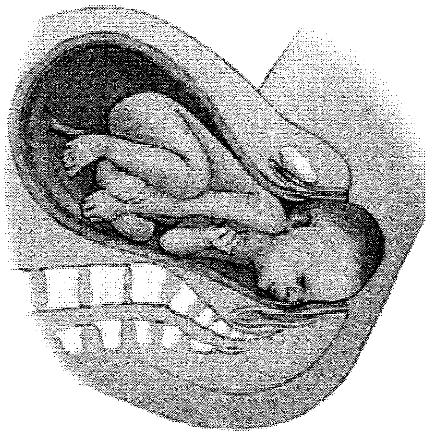

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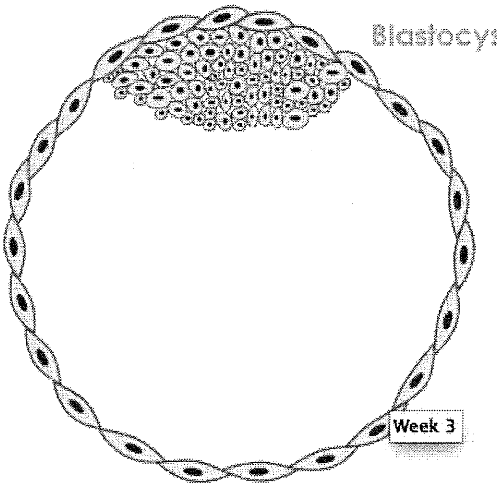
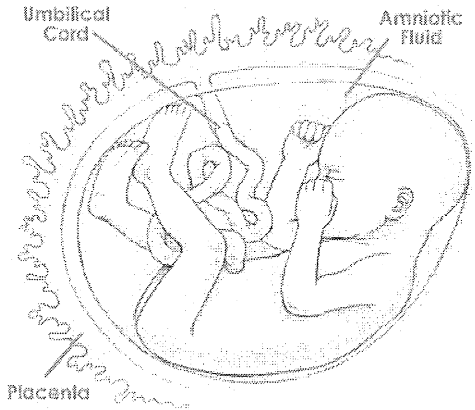


59. ______________________________



60. ______________________________

Label the following:

What is the difference between ultrasound and x-ray?

Why do we use ultrasound rather than x-ray to monitor the fetus' growth?

What do the following hormones do during pregnancy?

Hormone	Function
Oxytocin	
Progesterone	
Prolactin	
Human Chorionic Gonadotrophin	
Estrogen	

When do the following milestones occur?

a) Smiling
b) Laughing
c) Crawling
d) Sitting up
e) Start teething

Describe 2 different reflexes that newborns have